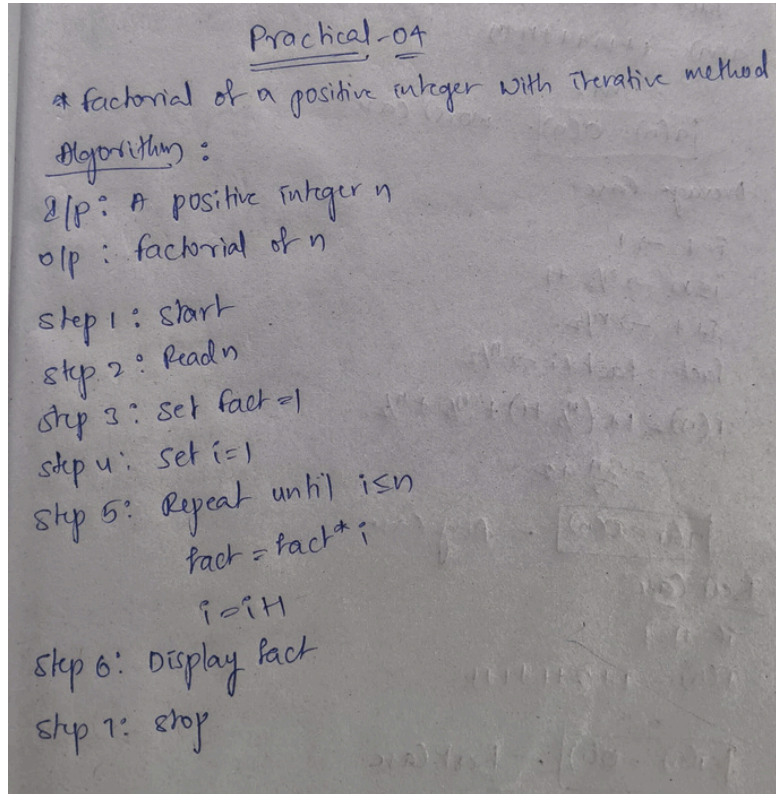


Practical- 4

Aim: Implementation and Timeanalysisoffactorialprogram using iterative and recursive method.

Iterative Method

Algorithm:



Program:

```
#include <iostream>
using namespace std;
class Factorial
{
public:
    int fact(int n)
    {
        int result = 1;

        for (int i = 1; i <= n; i++)
        {
            result = result * i;
        }

        return result;
    }
};
```

```
int main()
A.Abhilash
92400118815(5EN11)
```

```
{
    int n,result;
    Factorial obj;

    cout << "Enter the value to find its factorial: ";
    cin >> n;
    result = obj.fact(n);

    cout << "Factorial of " << n << " is " << result << endl;

    return 0;
}
```

Output:

```
PS C:\MinGW\bin> & "C:\Users\chinn\Downloads\dsa in c++\practical-4\factorial.exe"
Enter the value to find its factorial: 6
Factorial of 6 is 720
```

Time Complexity:

*Time complexity (iterative method):

```
for (i=1; i<=n; i++) {
    fact = fact * i;
}
```

$i=1 \rightarrow (1)$
 $i \leq n \rightarrow (n+1)$
 $i++ \rightarrow (n)$
 $fact = fact * i \rightarrow (n)$

$T(n) = 1 + n + 1 + n + n$
 $= 3n + 2$
 $T(n) = O(n)$ — worst case

Average Case

$i=1 \rightarrow 1$
 $i \leq n \rightarrow n/2 + 1$
 $i++ \rightarrow n/2$
 $fact = fact * i \rightarrow n/2$

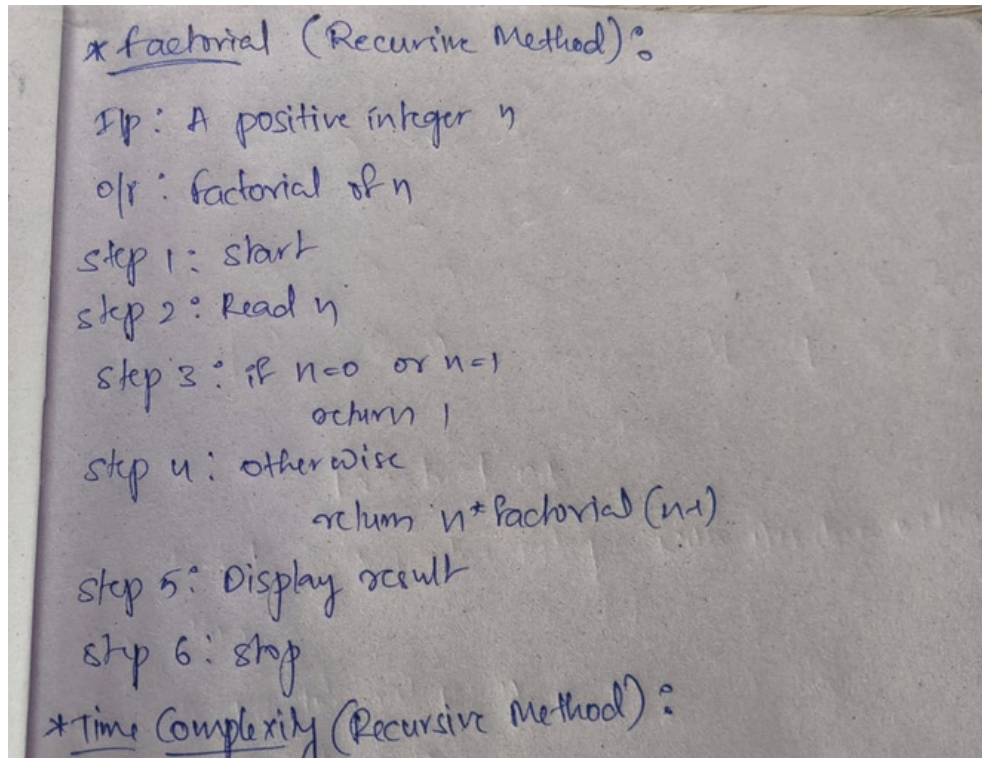
$T(n) = 1 + (n/2 + 1) + n/2 + n/2$
 $= \frac{3n}{2} + 2$
 $T(n) = O(n)$ — Avg case

Best Case:

if $n=1$
 $T(n) = 1 + 2 + 1 + 1$
 $= 5$
 $T(n) = O(1)$ — Best case

Recursive Method

Algorithm:



Program:

```
#include <iostream>
using namespace std;
class Factorial
{
public:

    int factr(int n)
    {
        if (n == 0 || n == 1)
        {
            return 1;
        }
        else
        {
            int result = n * factr(n - 1);
            return result;
        }
    }
};

int main()
{
    int n, result;
    Factorial obj;

    A.Abhilash
    92400118815(5EN11)
```

```
cout << "Enter the value to find its factorial: ";
cin >> n;

result = obj.factr(n);

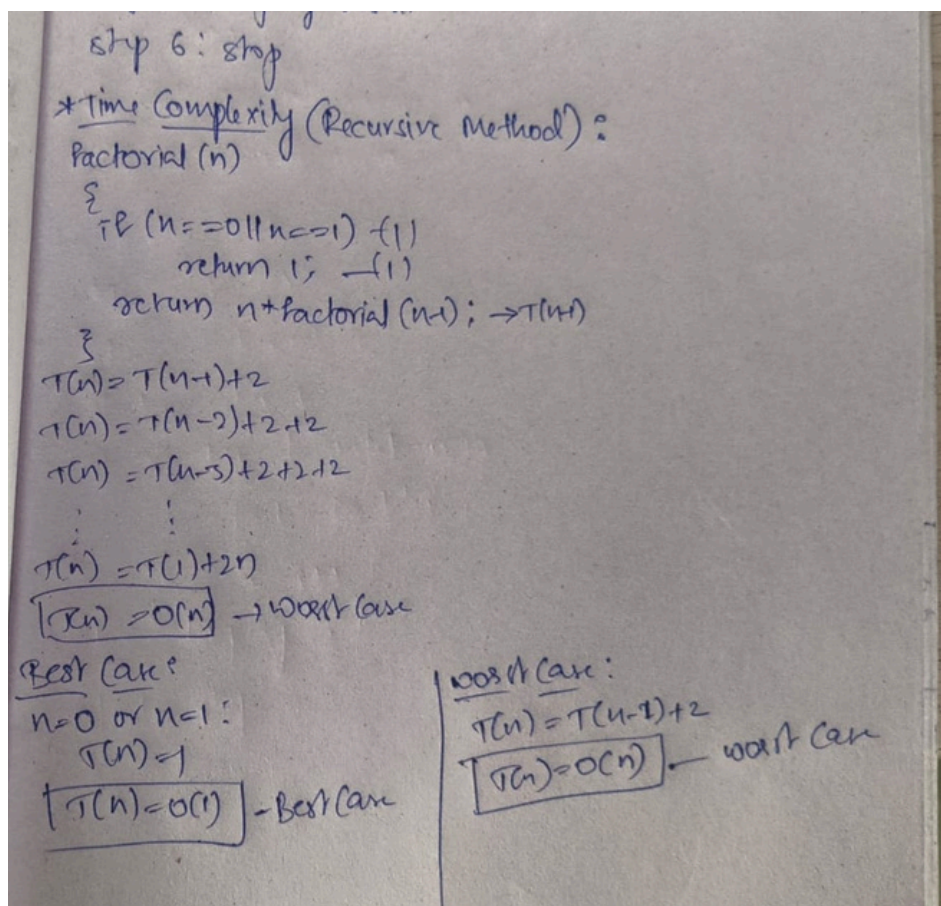
cout << "Factorial of " << n << " is " << result << endl;

return 0;
}
```

Output:

```
PS C:\mingw\bin> & "C:\Users\chinn\Downloads\dsa in c++\practical-4\recursive.exe"
Enter the value to find its factorial: 6
Factorial of 6 is 720
```

Time complexity:



Conclusion:

The factorial program was implemented using both iterative and recursive methods. The iterative method uses a loop, while the recursive method uses function calls. Both methods have $O(n)$ time complexity in the general case. The iterative method uses $O(1)$ space, whereas the recursive method uses $O(n)$ space due to the recursive call stack.